

Gossip-Scaled Semantic Forwarding in Named Data Networking

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Abstract—When a client asks for `/hospital/temp-sensor/room-101` and the network only advertises `/smart-hospital/building-a/floor-1/temperature/room-101`, exact-match forwarding drops the request. Embedding the name and matching it by cosine similarity fixes that, at the cost of running a sentence encoder in the forwarding path. Per-router caching does not amortise that cost: split the same traffic across more edge routers and each cache sees a thinner slice, so over a short horizon the network re-runs the encoder *more* as it grows. We share a resolution over anti-entropy gossip once, and only once a returned Data packet has proved it. Over a 60s run at 16 edge routers this holds the growth in encoder invocations to +9.9% against a per-router cache’s +59.6%, removing 27.7% of inferences; we also measure the two conditions that bound the claim, since the saving falls to 6.5% over 600s and reverses below about four edge routers. Two further results follow from the same verification signal. First, the feedback a forwarding plane calibrates against is *biased* rather than merely noisy: a producer’s refusal conflates “this name is not mine” with “this name is mine but not in that wording”, and we trace the conflation to a concrete protocol gap—NDN’s Nack has no equivalent of HTTP’s 406 or DNS’s NODATA. Second, because a gossiped mapping is a falsifiable claim, a router can audit its peers using verdicts it already collects; this attenuates a persistent poisoning attacker by a factor that *grows* with attack intensity, from $1.2\times$ at the lowest injection rate to $2.0\times$ at the highest. We report an attenuation, not a breakdown bound: the rate-invariance such a bound predicts was tested and did not hold.

Index Terms—Named Data Networking, semantic forwarding, anti-entropy gossip, conformal prediction, Byzantine resilience.

I. INTRODUCTION

Named Data Networking (NDN) [12], [13] forwards on names, and a name that does not match the Forwarding Information Base (FIB) is dropped. Real deployments break on this: two organisations naming the same sensor differently cannot reach each other’s data even when both are reachable. Amadeo *et al.* [1] answer it by embedding names and forwarding on cosine similarity, caching each resolution in an *Embedding Store* at the router that made it.

That design leaves a scaling problem its own conclusion names as future work. A per-router cache is populated by the traffic that router sees. Add edge routers without adding traffic and every cache thins, so the network pays for more encoder invocations in total. Our measurements put the effect at +59.6% more inferences from 1 to 16 edge routers on one catalog and +67.4% on the other, for identical offered load.

This paper’s premise is that the fix is not a bigger cache but a shared one, and that NDN already supplies what makes

sharing safe. A semantic match is a guess; a returned Data packet proves it. Gossiping only proven mappings means a wrong guess never propagates, which is what lets the similarity threshold be permissive without the usual cost.

Contributions. We claim three things, each scoped to what we measured.

- 1) *Sharing, not caching, is what keeps recognition cost from growing with the network* (§IV). We also measure the two conditions that bound this: the advantage decays with run length, and reverses below about four edge routers.
- 2) *The free labels a forwarding plane calibrates on are biased* (§VI), and the bias has a nameable protocol cause rather than being noise. We found and corrected an instance of it in our own producer model, and report the regime in which correcting it helps and the regime in which it does not.
- 3) *A verifying forwarding plane can audit its own peers* (§VII). Byzantine-robust gossip normally cannot detect a liar and must settle for bounding its influence; a gossiped mapping is falsifiable, so detection is available from verdicts already collected.

What we do not claim. Edge tagging—annotating an Interest with the resolved prefix so later hops skip the encoder—is Amadeo *et al.*’s mechanism, inherited here. Anti-entropy dataset synchronisation is a decade old in NDN [5], [6]; our gossip layer is that mechanism carrying a different payload. Learning per-item decision boundaries under an error budget is what vCache [7] does for LLM prompt caching. We evaluated an error budget as a replacement for the similarity threshold, it lost to a threshold tuned on one catalog and carried to another, and we report that as a withdrawn claim (§V).

Organisation. Section II places this work against semantic forwarding in ICN, the synchronisation layers NDN already provides, and the literature on learned decision boundaries and Byzantine-robust gossip. Section III describes the simulator, the producer model and the measured cost basis. Section IV presents the scaling result, the two conditions bounding it, and the gossip-period frontier. Section V covers calibration from free labels and the claim we withdrew. Section VI establishes that those labels are biased and locates the protocol gap responsible. Section VII gives the threat model and the reputation mechanism built on verification. Section VIII reports latency, energy, encoder choice and schema drift; Section IX states the limitations, and Section X concludes.

II. BACKGROUND AND RELATED WORK

Semantic forwarding in ICN. Chan *et al.*’s Fuzzy Interest Forwarding [2] matches name components with Word2Vec at both Content Store and FIB, and is the original statement of the problem. Amadeo *et al.* [1] use a sentence encoder, settle on a threshold of 0.7, and evaluate a single access router. Raza *et al.*’s INF-NDN [3] routes LDA topic tags through central supernodes; it is a *competing* answer to the problem we attack, concentrating semantic resolution rather than replicating or sharing it.

A note on “SAF”. In NDN, SAF already denotes Stochastic Adaptive Forwarding [4]. We write “semantic forwarding” or name the authors; our result files retain the key `saf` for continuity.

Synchronisation NDN already has. The anti-entropy and rumour-spreading pair we use is the classical epidemic construction of Demers *et al.* [16], whose $O(\log N)$ propagation under random peer choice [17] is what §fsec:frontier preserves by keeping one uniform pick. ChronoSync [5] and its successors reconcile datasets by exchanging condensed digests, and NLSR [6] already floods name-prefix reachability over such a layer; Moll *et al.* [18] survey how these designs differ. Our contribution is not the mechanism but the payload and the precondition: a wording-to-route mapping, shared only after a Data packet proved it. §IV measures an approximation of an unbounded sync layer precisely because “why not just announce the prefix” is the first objection this design invites.

Learned boundaries under an error bound. vCache [7] learns per-item thresholds under a user-set error bound for prompt caching; Finamore *et al.* [8] do error-controlled similarity caching in a networking setting. The boundaries themselves are conformal in the sense of Vovk *et al.* [19]; pooling calibration across parties is federated conformal prediction [9], and the drifting variant we evaluate follows adaptive conformal inference [20]. That a threshold tuned on one catalog need not hold on another is the covariate-shift problem Tibshirani *et al.* [21] treat formally. What is specific to NDN is that the labels are free: an LLM cache needs a judge model to decide whether a hit was correct, which is the expense one was trying to avoid, whereas a forwarding plane receives Data or a Nack either way.

Biased feedback. §VI is an instance of missing-not-at-random feedback, the pattern Saito *et al.* [10] treat in implicit-feedback recommendation: the label a route receives depends on a producer’s declared vocabulary, which is also what determines whether the request could have been served. Our contribution is locating it in a forwarding plane and identifying the protocol gap that causes it.

Byzantine-robust gossip. Robust-aggregation methods bound an adversary’s influence—clipping, trimming, bounded contribution [11]—because in gossip learning a neighbour ships a gradient no honest node can check. §VII exploits the fact that a gossiped *mapping* can be checked.

III. SYSTEM MODEL AND METHOD

A. Simulator

Names are embedded with a MiniLM sentence encoder [15], a distilled model in the Sentence-BERT family [14] and the one the baseline settles on. We use a discrete-event model in which each router has a single-server FIFO queue for packet processing, so a strategy that adds work to the forwarding path pays for it in queueing delay rather than in a constant read from a configuration file. A forwarding strategy is consulted at exactly one point—the FIB miss—and told the outcome afterwards. Content Store, PIT and exact FIB lookup are identical across strategies, so a difference in results is a difference in policy.

Similarity is permitted to choose a *route* and never to decide that two names denote the same object: the Content Store and PIT remain strict exact-match. A semantic hit in the Content Store would hand a client data it did not request; a semantic hit in the PIT would return data down the wrong chain.

B. Costs are measured, not assumed

Every latency constant is measured on the machine running the study and recorded with a flag saying whether it was measured or assumed. One MiniLM-L6 inference costs 7.045 ms. A cosine search over the 50-entry FIB costs 0.0045 ms. The ratio is about 1566:1, so *the encoder is the only cost that matters*: any scheme that optimises the similarity *search* optimises a rounding error. This is why our mechanism targets the number of inferences rather than their individual cost.

C. Producers are modelled, not oracular

The weakest assumption available here would be to let a producer consult ground truth when deciding whether a semantically-resolved Interest was meant for it. Instead each producer holds a *declared schema*: the instance it is attached to and the metric terms it answers to, exactly as a service registry or an OpenAPI description would provide. Declarations are deliberately incomplete; `alias_coverage` controls how incomplete. At coverage 0.7 a producer leaves 29% of the wordings it can be asked by undeclared, and refuses requests genuinely meant for it. The feedback channel is therefore informative rather than correct, which is the situation any deployment is in.

D. Workload and metrics

Requests are Poisson arrivals over a Zipf ($\alpha = 1.2$) popularity distribution, with each consumer population drawing from an overlapping but distinct slice of the catalog—without that overlap parameter, sharing would be either trivially redundant or trivially useless. Each catalog carries 50 services, 300 rewordings and 350 distractors that no service can satisfy, so precision is measurable.

Scoring is correctness-aware. Every Interest ends in exactly one of *correct*, *misdelivered*, *rejected* or *timeout*, compared against ground truth the routers never see. A single accuracy number would reward a strategy that answers nothing, since

distractors are supposed to be rejected, so precision, recall and false-positive rate are reported separately.

E. Comparison basis

Because the baseline is what this work extends, we reproduce its evaluation setting rather than inventing one. The access-edge topology, the one-way latencies (10 ms client-to-router, 20 ms router-to-provider), the FIB of 50 entries, the request rates of 30–300 Interests/s, the Zipf skew of 1.2, a disabled Content Store and an unbounded PIT are all taken from Amadeo *et al.* [1]. Our catalogs share their construction exactly: 50 services, each with one canonical name and six rewordings, giving 350 satisfiable names, plus an equal number of unrelated names—their harder configuration, in which half the offered catalog cannot be served by anything.

We report their metrics. *Service discovery time* is our Interest-satisfaction time and decomposes identically into client-to-router, queueing, processing and router-to-provider components. *Processing time* is our encoder invocation count multiplied by the measured inference cost, plus queueing. Their *share of Interests forwarded successfully on the first attempt* is our satisfaction rate, with one deliberate difference stated plainly: they count an Interest as successful when it is forwarded and answered, while we additionally require the Data to have come from the service the name should resolve to. A confident match to the wrong room’s sensor is a success under the first definition and a misdelivery under ours. We use the stricter one because a forwarding plane that resolves names semantically can fail in exactly that way, and a metric that cannot see the failure cannot be used to tune against it.

Absolute numbers are therefore not comparable across the two papers, and we do not compare them. Amadeo *et al.* report essentially complete forwarding at $\text{Th} = 0.7$ on their catalog; at the same threshold ours yields recall of 0.866 and 0.804. The gap is a property of the rewordings, not of the method: their variations are synonym substitutions, whereas our six rewrite families also drop levels of hierarchy, reorder components and abbreviate them, and a seventh family draws terms from published IoT vocabularies. Ours is a harder instance of the same catalog design. Every comparison in this paper is therefore between strategies measured inside one harness, never between our numbers and a published one.

F. Cross-validation

A hand-written simulator can be perfectly self-consistent and still model NDN incorrectly. We cross-check the transport layer—the part that is *not* our contribution—against ndnSIM on the same topology, link delays and request pattern with semantic resolution disabled. Median round-trip times agree to within 0.11 ms once the 1 ms producer service time our model charges is accounted for.

All results are 20 seeded runs reported with a 95% confidence interval. Runs are reproducible across processes: three separate defects in which an interpreter-salted iteration order reached the event queue were found and fixed during develop-

TABLE I
ENCODER INVOCATIONS AGAINST EDGE ROUTERS, 60 s, IDENTICAL OFFERED LOAD. GROWTH IS FROM 1 TO 16 EDGE ROUTERS.

	1	4	8	16	growth
<i>hospital</i>					
Semantic, no cache	2022	2062	2073	2079	+2.8%
Per-router cache	833	1006	1155	1330	+59.6%
GS-NDN (gossiped)	875	881	918	962	+9.9%
<i>city</i>					
Semantic, no cache	2017	2060	2072	2079	+3.1%
Per-router cache	764	951	1102	1278	+67.4%
GS-NDN (gossiped)	781	812	856	916	+17.3%

ment, and a regression test executes two child processes under differing PYTHONHASHSEED values and compares metrics.

IV. SCALING: SHARING VERSUS CACHING

Table I is the primary result. Semantic forwarding without a cache pays for every FIB miss, so it has nothing to erode and barely grows. The per-router cache thins as routers multiply. Sharing holds growth to roughly a sixth of that, removing 27.7% of inferences at 16 edge routers on hospital and 28.3% on city. The claim is independent of any threshold or error budget—it is a property of how many times the encoder runs.

A. Two conditions that bound it

The horizon. A cold cache costs one inference per router per wording: N routers pay it N times, but they pay it once. Over 600 s instead of 60 s the per-router cache grows +14.4% rather than +55.2%, and gossip’s saving at 16 edge routers falls from 25.6% to 6.5% (city: 26.8% to 9.2%). The ordering never reverses, but the magnitude does. Sharing pays a network that is large and still learning its catalog.

Network size. At a single edge router there is nobody to share with, and anti-entropy costs 4–5% more inferences than a plain per-router cache. The benefit appears from about four edge routers upward.

B. Is the decay an artifact of a closed catalog?

A natural hypothesis is that the horizon decay is an artifact of our catalog being finite: once every router has met all 300 rewordings there is nothing left to learn. We tested it by introducing wordings over the run instead of making all of them askable at $t = 0$, drawn from an independent random stream so that only *which* wording a request uses changes.

The hypothesis is false, and instructively so. A slower arrival rate lowers the saving at *every* horizon rather than sustaining it: from 18.0% (closed) to 2.5% (one new wording every 5 s) at 60 s, and the ordering is monotone across the whole grid. Gossip’s saving comes from the *backlog* of resolutions some routers hold and others do not, and a closed catalog at $t = 0$ is the largest backlog obtainable. The decay is real and §IV’s scope condition stands as stated.

TABLE II
ANTI-ENTROPY PERIOD AT 16 EDGE ROUTERS: INFERENCE SAVING
AGAINST GOSSIP BYTES. UNIFORM PEER SELECTION VERSUS SELECTION
BIASED BY PENDING DELTA.

period	uniform		delta-biased	
	saving	kB	saving	kB
0.25 s	24.4%	933	24.2%	919
0.50 s	25.6%	746	26.1%	718
1.00 s	24.7%	707	25.8%	685
2.00 s	21.5%	603	21.7%	606
5.00 s	18.0%	433	18.0%	432

C. Would a more aggressive sync layer do better?

NDN already ships anti-entropy synchronisation, so the obvious objection is to turn it on and drop this layer. We bound that with an arm in which every router reconciles with every peer each round with no cap per exchange. It removes *fewer* inferences than the bounded protocol (21.0% against 25.6% at 16 edge routers) while sending 1.2–1.6× the bytes. Anti-entropy at fanout 2 already reaches everyone in $O(\log N)$ rounds; what limits the saving is how many distinct wordings the network has yet to see, which no amount of synchronisation changes. The part of the design doing the work is *what* is shared and *when*—a mapping, only once a Data packet proved it—not how aggressively.

D. A Pareto frontier, not an operating point

The gossip period is a degree of freedom, and reporting a single value would hide a trade the reader should see. Table II sweeps it. Two observations matter.

First, the exchange rate between bytes and inferences depends on network size. At 4 edge routers the saving is flat across the whole range (8.5–9.6%) and a longer period is nearly free; at 16 edge routers moving from 0.5 s to 5 s costs 7.6 points of saving to halve the bytes. A conclusion drawn at one operating point would have been wrong at the other. Second, 0.25 s is worse than 0.5 s on *both* axes, so the chosen default is Pareto-optimal among measured points rather than conventional.

Delta-biased peer selection. Once the network converges, the modal anti-entropy exchange is two agreeing digests and no payload. Ranking candidate partners by how far behind they are—computable from the watermark the sender already keeps, requiring no messages and asking the peer nothing—redirects those exchanges. One pick out of *fanout* remains uniform: a purely greedy rule is no longer an epidemic process and can starve a peer that never tops the ranking, trading bounded propagation for unbounded.

This is the only mechanism we tested that moves the frontier outward rather than sliding along it. Paired seed by seed at 16 edge routers it saves 11.1 ± 7.5 additional inferences while sending $3.68 \pm 0.96\%$ fewer bytes at 0.5 s (hospital, 14/20 and 19/20 seeds), and 24.9 ± 7.3 inferences at 1.0 s (20/20 seeds). Satisfaction is unchanged, which is correct for a mechanism that redirects exchanges rather than changing

what is forwarded. The honest magnitude is 1–1.5% of encoder invocations and 3% of bytes: a refinement that costs nothing, not a headline. It helps only in the middle of the period range—at 0.25 s everyone is already current and at 5 s rumour push carries almost everything—which is itself evidence that it works for the stated reason.

V. CALIBRATING FROM FREE LABELS

Amadeo *et al.* select a threshold of 0.7 on their own catalog. On ours that setting yields recall of 0.866 and 0.804 while the best operating points sit nearer 0.5 and 0.45. A threshold is a property of the catalog it was tuned on. Worse, one number must serve every route: some services sit alone in embedding space and a loose score is safe, others have near-identical siblings and a high score is still a coin flip.

Every semantic forwarding decision is an experiment the network answers, so (score, verdict) pairs accumulate for free. For route r with observations (s_i, y_i) we take the boundary τ_r as the smallest score such that, among observations at or above it, the upper confidence bound on the error rate stays within an operator-set budget ε :

$$\tau_r = \min\{s : \text{UCB}_\delta(\hat{e}_{\geq s}, n_{\geq s}) \leq \varepsilon\} \quad (1)$$

using a Wilson bound, because with a dozen observations the normal approximation assigns a route with no errors a bound of exactly zero. Routes are estimated hierarchically, shrinking toward a pooled boundary at a rate set by their own evidence: a single route is seen a few dozen times at most, and a conservative bound on that many observations is nearly vacuous.

Risk control deadlocks without exploration. A boundary set too high blocks exactly the decisions that would produce the evidence to lower it. Left unexplored the controller settles at refuse-everything and stays there: at $\varepsilon = 0.02$ it reaches 0.732 satisfaction, and tightening ε further changes nothing because it is already refusing what it can. Spending 5% of refused decisions on evidence reaches 0.818. This is not an estimator bug but a property of learning from one’s own choices.

The efficiency claim was tested and withdrawn. The budget holds out-of-sample at every setting we tried—realised error stays under ε in all fourteen tests. But against a fixed threshold tuned on one catalog and carried to the other, the risk controller Pareto-dominates in *none* of fourteen comparisons, while the transferred threshold dominates in seven. What survives is narrower and we report it as such: the controller needs no labelled sample of the domain it runs on, because it calibrates from live producer feedback. An operator who *has* a labelled target-domain catalog should tune a threshold on it and will do slightly better.

VI. THE FEEDBACK IS BIASED, NOT JUST NOISY

This is the finding the paper stands on alongside §IV, and it emerged from a defect in our own producer model.

A producer refusing a semantically-resolved Interest has two distinct reasons, and NDN’s Nack cannot distinguish them.

This name is not mine is a routing fact: the mapping that sent the Interest here is wrong. *This name is mine but not in that wording* is not: the route was correct and only the phrasing fell outside what the producer declared. Collapsing the two makes every gap in a producer’s vocabulary look like evidence against a route that was in fact right. HTTP separates 404 from 406 and DNS separates NXDOMAIN from NODATA for this reason; semantic forwarding is what makes the distinction necessary in NDN.

The distinction must be one the producer can actually make. Our first implementation tested whether the producer published *the prefix the router chose*—a test a misroute satisfies by construction, since that is why the Interest was routed there. Measured at full declaration coverage, where a genuine wording gap is nearly impossible, 132 of 157 refusals reported “unknown wording” about a route that was simply wrong, 2 were the case the label exists for, and the routing-error label never fired once. The channel carried almost no information.

What a producer genuinely knows is which *instance* it is attached to—the room, the junction, the meter. A request that does not name that instance is not for this service whatever the wording. That test is exact in our runs: every instance mismatch was a real misroute or an unsatisfiable distractor, with no false positives. It converts a dead channel into an informative one: at coverage 0.5, 291 refusals are correctly labelled routing errors and 546 are wording gaps, 71% of them genuine.

The residue is irreducible, and it inverts. A producer serving temperature in room 101, asked for humidity in room 101, cannot distinguish a metric it does not serve from a synonym of its own it forgot to declare. The base rate of genuine gaps among such refusals moves from 0.021 at full coverage to 0.711 at half coverage—and the right policy inverts with it. Ignoring ambiguous refusals gains 0.07 satisfaction at coverage 0.5 (0.844 against 0.774) and costs 0.014 at full coverage on city (0.963 against 0.978). Declaration coverage is a property of other organisations’ service registries, which an operator does not observe.

We attempted to close this with inverse propensity weighting, estimating the propensity per route. It failed, and the failure is informative: the estimated weight sat between 0.414 and 0.470 in *every* regime and both catalogs, because a correct route at low coverage both serves often and is refused often, so a route-conditioned ratio barely moves. The quantity that does move is a single network-wide base rate. We report the regime inversion as an open problem with its magnitude measured, rather than as a mechanism that closes it.

VII. THREAT MODEL AND PEER REPUTATION

A. Threat model

What the attacker controls. A fixed share of routers, chosen uniformly at random. A compromised router is a full participant holding real keys and peer relationships, so nothing here is defeated by authenticating the channel. It behaves correctly on the forwarding path and lies only into gossip—a router that drops traffic is a denial-of-service problem the

TABLE III
SATISFACTION AND REALISED ERROR AGAINST COMPROMISED SHARE, 8 EDGE ROUTERS. “IMPORTS UNVERIFIED” IS THE BEHAVIOUR BEFORE WE EXTENDED VERIFICATION TO MAPPINGS LEARNED OVER GOSSIP.

compromised	0%	12.5%	25%	50%
<i>hospital</i> , satisfaction				
Imports unverified	0.940	0.832	0.784	0.747
GS-NDN	0.941	0.894	0.872	0.854
+ reputation	0.941	0.930	0.922	0.909
<i>hospital</i> , realised error				
Imports unverified	0.027	0.103	0.139	0.164
GS-NDN	0.026	0.058	0.070	0.082
+ reputation	0.026	0.035	0.037	0.049
<i>city</i> , satisfaction				
GS-NDN	0.960	0.912	0.890	0.871
+ reputation	0.960	0.949	0.941	0.930
<i>city</i> , realised error				
GS-NDN	0.007	0.044	0.060	0.071
+ reputation	0.007	0.015	0.021	0.034

ICN literature already treats [22], and it would confound the measurement of what the learning channel is worth.

What it can do. Inject fabricated mappings and fabricated calibration evidence, re-injecting both every round so that retraction after one round trip settles nothing. It knows the catalog well enough to target names clients actually request, which is the strong assumption here.

What it cannot do. Forge another router’s identity, partition the network, observe traffic it does not carry, or make a producer answer a name it does not publish. This boundary matters for §VI: the refusal reason is authored by the producer, so a compromised *router* cannot forge one, and what a compromised *producer* could do is unmeasured.

What limits damage structurally. A router’s own confirmed mappings outrank anything it is told. That is not a defence added against this attack; it falls out of gossiping only what a returned Data packet proved.

B. Verification-grounded reputation

Byzantine-robust gossip normally defends without ground truth and must bound an adversary’s influence rather than identify it. A gossiped mapping is different: it is a falsifiable claim, and the producer answers it. We therefore charge each refusal to the peer that supplied the mapping, and never reinstall a pair the router disproved first-hand. Trust is a Wilson *lower* bound on a peer’s success rate—mirroring the upper bound on error in §V, both bounding in the direction that costs us. A peer with no history is believed, deliberately: distrusting unaudited peers would break the bootstrap gossip exists for.

Calibration evidence gets a different defence, because an observation about a route is not immediately falsifiable the way a mapping is. No single peer may occupy more than a quarter of a route’s calibration window, which bounds contribution by construction rather than by detection.

Table III reports the effect. At 50% compromise reputation recovers 0.162 of the 0.193 satisfaction the attack costs on

hospital (84%) and 0.177 of 0.207 on city (86%), and cuts realised error by 40% and 52%. It is free when nobody is attacking—0.941 against 0.941—which is the number that decides whether it can be left switched on. It works for the stated reason: the count of distrusted peers rises linearly with the compromised share, reaching 29.3 at 50%.

C. An attenuation, not a breakdown bound

It is tempting to formalise the residual as a bound. Each pair of an honest router and a compromised neighbour admits at most k mappings before the liar is cut off, where k is the trust prior, giving admitted poison $\lesssim 2Ef(1-f)k$ for E adjacencies and compromised fraction f —quadratic in f , and *independent of injection rate*.

We tested both halves. The mechanism half holds: sweeping k over $\{1, 2, 4, 8, 16\}$ moves surviving poison monotonically from 494 to 1067, so the admission fee is real and is the knob. **Rate invariance fails.** Taking the attacker from 1 to 16 fabricated mappings per round raises the defended arm’s realised error by $1.43\times$ on hospital and $1.91\times$ on city. A bound counted in pairs does not move with rate; this moves. We therefore do not claim a breakdown point.

What survives is an attenuation whose value *grows* with attack intensity. The undefended arm rises $2.35\times$ and $3.25\times$ over the same range, so the ratio between defended and undefended widens from $1.24\times$ at the lowest rate to $2.04\times$ at the highest on hospital, and $1.72\times$ to $2.93\times$ on city. The defence is worth more the harder it is attacked.

The sweep also selects a better prior than intuition did. A fee of $k = 1$ is best under attack but falsely distrusts 10.9 honest peers on a clean network and degrades sharing by 3%; $k = 2$ distrusts 1.1 for 0.06%, while realising 0.037 error against $k = 4$ ’s 0.044. We use $k = 2$. The honest-network control is what makes that visible—the attack column alone would have argued for 1.

VIII. SUPPORTING RESULTS

Latency under load (Fig. 3). Encoding on every FIB miss saturates the forwarding path: p95 round-trip time rises from 79 to 172 ms between 30 and 300 Interests/s, while cached and gossiped variants stay flat at 78–83 ms. The knee sits just past 200 Interests/s, which independently reproduces the bottleneck Amadeo *et al.* report beyond roughly 210 Interests/s for the same encoder—a point of agreement we did not tune for and that we take as evidence the cost model is calibrated correctly.

Energy. On the radio model of Askar *et al.* [23], with encoder energy charged at measured CPU draw, GS-NDN uses 634 J against 735 J for uncached semantic forwarding and 654 J for a per-router cache, at higher satisfaction.

Does the transformer earn its cost? A character n -gram control reaches 0.791 satisfaction where MiniLM reaches 0.969 at the same threshold. It resolves abbreviations and fails on true synonymy, which is the gap the encoder is paying for.

Schema drift. Producer relocation hurts every strategy equally—FIB withdrawal invalidates a stale mapping before any producer can refuse it. Schema drift, where a producer

quietly narrows what it answers to without any routing event, removes that confound, and there verification’s advantage appears: $+0.0042 \pm 0.0018$ satisfaction on hospital (16/20 seeds) and $+0.0124 \pm 0.0045$ on city (19/20), paired seed by seed. Nothing in the routing plane observes a schema drift; feedback is the only signal that can.

IX. LIMITATIONS

Our synonyms are easier than standardised ones. The catalogs are generated from a lexicon we wrote. To test whether that lexicon flatters the encoder, we added rewordings transcribed from Brick, SAREF, Project Haystack and W3C SSN/SOSA. They are markedly harder: on city, rank-1 recognition is 0.578 for ontology-derived wordings against 0.920 for ours. This *confirms* the concern rather than removing it, and every recognition number here should be read as an upper bound.

Provenance is not implemented. Signing an observation so a lie can be attributed is the other defence our threat model names, and it is absent. A deployment facing this threat model needs it; we show the attack lands and by how much, not that reputation alone suffices.

A compromised producer is unmeasured. Producers are trusted about their own services throughout.

The encoder guard is a trade, not a win. Refusing to pool calibration scores across routers running different encoders yields lower satisfaction and lower realised error than pooling them naively (0.948/0.023 against 0.951/0.027 at a quarter heterogeneous). For an error-budget system that trade is defensible, but it is not a clean improvement.

X. CONCLUSION

Sharing a verified resolution by gossip, rather than caching it per router, keeps recognition cost from growing with the network—by an amount we bound with the two conditions we measured rather than assumed. The same verification signal that makes sharing safe also lets a forwarding plane generate its own calibration labels and audit its own peers. Those labels are biased in a way we traced to a missing distinction in NDN’s Nack, and the regime in which correcting that bias helps inverts on a quantity operators cannot observe. We report that as an open problem with its magnitude measured, together with two mechanisms that did not survive their own evaluation, because a design space is described as much by what fails in it as by what works.

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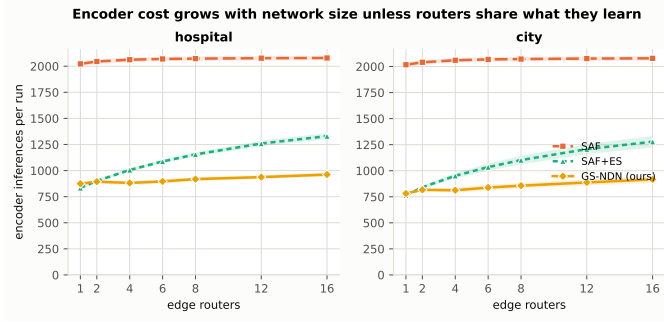


Fig. 1. Encoder invocations against edge routers. The per-router cache grows with the network; gossiped sharing stays nearly flat.

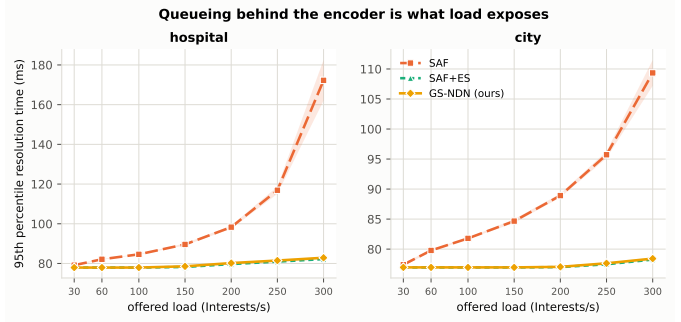


Fig. 3. p95 round-trip time against offered load at a single edge router.

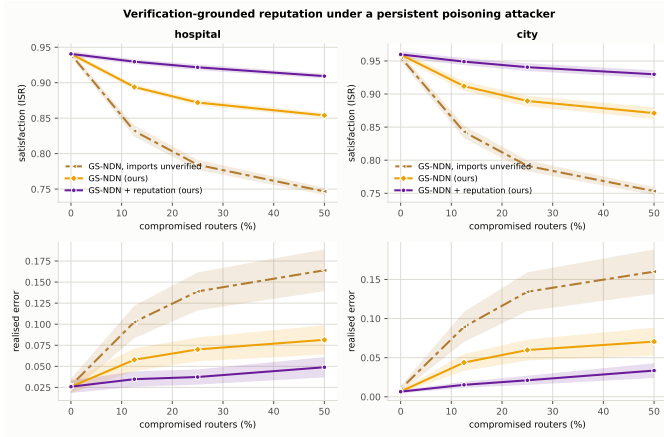


Fig. 2. Satisfaction and realised error under a persistent poisoning attacker. Reputation is free at zero compromise and recovers most of the attack's cost thereafter.

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